Quantification of seismic performance factors of Hexagrid, Diagrid and tube structural systems for tall buildings

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ABSTRACT

Vertical growth of cities, has become a challenging issue in the building industry, due to the heavy urbanization and population growth. Bracing, is a highly efficient and economical method in order to achieve resistance against lateral loads in a tall building with steel structure. In order to improve the structural efficiency of tube-type skeletons in tall buildings, a new structural system, known as hexagrid, is investigated in this paper. In comparison with the diagrid system, the hexagrid, consists of multiple hexagonal grids on the periphery of the building. This investigation quantifies three seismic performance factors, including over-strength factor (Ω_0), period-based ductility (μ_T) and response modification coefficient (R) for hexagrid, diagrid and tube systems. The present study involves the 30- and 50-story of peripheral hexagrid and diagrid bracing systems compared with the tubular system. The structures were evaluated using nonlinear static and dynamic analyses. According to the results, the hexagrid system has a better response modification coefficient (R). Nonlinear static and IDA analyses are conducted using Open Sees software. The structures' behavior under 7pairs of ground motion records have been investigated by nonlinear time history analysis utilizing PERFORM-3D software. The maximum inter-story drift for the studied models was determined and compared with the "life safety" and "collapse prevention" performance limits, as recommended by FEMA P695.
Keywords: Hexagrid systems, Diagrid Systems, High-rise buildings, nonlinear analysis, inter-story drift,

1. INTRODUCTION

Construction of high-rise buildings used to be driven by the demand for space in densely populated areas. However, as the height of the building increases, the building becomes a symbol of prominence. Advancements in structural engineering and technology have greatly pushed the height limit. The growth was propelled by the invention of the safety mechanism for vertical elevators by Elisha G. Otis in mid-1800, which made the elevator the fastest and efficient means of vertical transportation in tall buildings. Combined with the improvement in fabrication and construction methods, the construction of skyscrapers not only has become more relevant and feasible; it has pushed the height limit even further.

The most profound advancements in structural engineering is the development of different structural systems that allow for taller buildings. As the height of the building increase, the lateral load resisting system becomes more important than the structural system which resists the gravity loads. The lateral load resisting systems that are widely used are the followings: rigid frames, braced frames, belt and outrigger truss systems and framed tube structures. A core or a system of cores which provides additional stiffness to the structure, is usually present in the building's structural system and uses for utilities such as the elevator shaft. A combination of the lateral load resisting systems or a variation of the concepts such as the braced tube system or the bundled tube system has resulted in emergence of high-rises with aspect ratio of 1:7 or even more.

The most significant trend of tall building configuration constructed in the late 19th century was based on the economic equations by introducing as much natural light as possible to have better views and occupant comfort requirements. In order to achieve this purpose, a new technology was developed which improved the structural systems [17].

It is required that the structural systems be in consistence with the architectural demands. Structure/architecture interaction motivates structural engineers to create new structural systems according to architectural requirements. Amongst all the popular structural systems, the most efficient ones are the ones with tube-type performance. In fact, locating a major part of lateral load-resisting system at the building's periphery has led to an efficient structural performance [13]

Nowadays, the diagrid structural system with tubular behavior is being employed in tall buildings as structurally efficient as well as architecturally satisfying structural system for tall buildings. Perimeter diagonals act as the façade, which improves the esthetics of the building to a great degree. The use of perimeter diagonals on the façade have caused it to densely concentrate the steel material and, due to its structural effectiveness, may architecturally weaken its function [13].

In recent years, some high-rises have been constructed with triangulated exterior structural members. This system is known as diagrid system. It is architecturally very appealing as shown by the Swiss Re Building in London and Hearst Tower in New York.

The present investigation aims to derive code-compliant seismic design factors for hexagrid, diagrid and tube systems through collapse based procedure. For this purpose, the seismic performance factors of the systems, including response modification coefficient (R), over-strength factor (Ω_0) and period-based ductility (μ_T), are quantified using the collapse-based methodology introduced by FEMA P695 [6]. In the following section, the considered buildings are designed by trial response modification coefficient (R). Two-dimensional nonlinear static analysis and incremental dynamic analysis (IDA) are performed to determine the Ω_0 and μ_T factors and verify the assumed R coefficient.

2. HEXAGRID AND DIAGRID PATTERNS

In order to evaluate the geometry, the mechanical behavior and the structural efficiency of the considered patterns, three distinct unitary "entitles", have been considered: (i) the module, (ii) the unit cell and (iii) and the representative volume element [16].

The module is the frame shape, i.e. the geometrical arrangement of the structural members giving a visual representation of the pattern (hexagonal for hexagrid and triangle for diagrid, *Figure 1*); anyway the replication of the module gives rise to overlaps of the edges so that the overall geometry cannot be obtained by simply copying the module. For this reason, it is necessary to identify the unit cell, defined as the geometric unit that, through replication, allows to obtain the overall geometrical pattern without overlaps or gaps (*Figure 1*).

While the unit cells represents the repetitive unit from the geometric point of view, the representative volume element, represents the structural idealization of the unit cell, which only can be established by anticipating the deformation modes and internal forces arising in the unit cell as a part of the global grid. For more information the reader is referred to Montouri [16] paper.

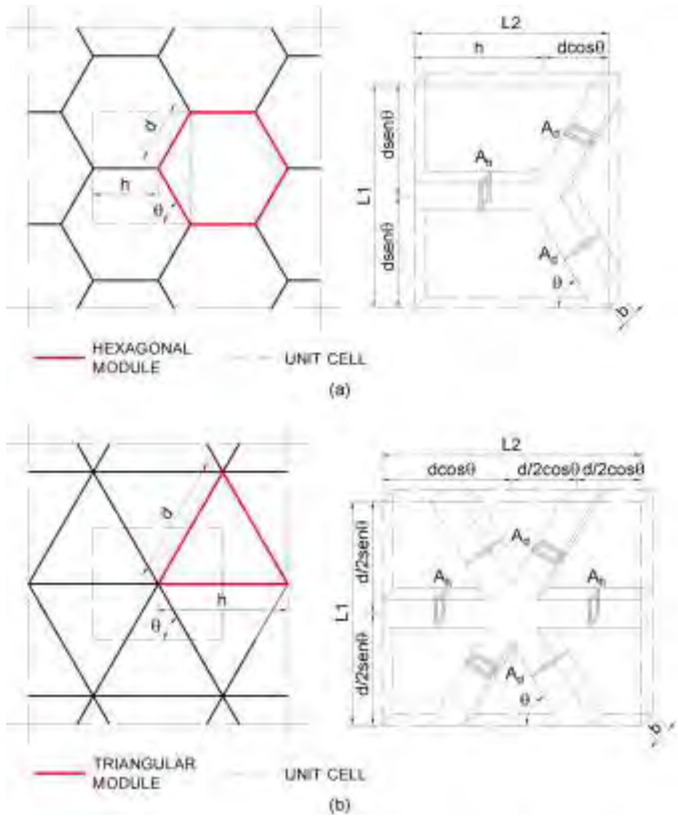

Figure 1. *Modules, unit cells and volume occupied by solid material in the unit cell, respectively for (a) horizontal hexagrid and (b) diagrid (Montuori, Fadda et al. 2015)*

3. STRUCTURAL MODELS AND DESIGN CONSIDERATION

The structural models in this research consists of three steel tall buildings with hexagonal, diagrid and tube structural skeleton in 30- and 50-story forms. The prototype buildings designed for high seismicity (i.e. PGA = 0.35 g) in accordance with current American seismic prescriptions [1]. Soil class C (i.e. $180\ m/s \le V_s \le 360\ m/s$) is assumed to perform the design by a series of response spectrum analysis on three-dimensional models (see *Figure 2*), including vertical seismic component, second order effects and accidental eccentricity of the seismic masses. The other key assumption made in trial design of buildings, is the response modification coefficient (R). Within ASCE 7-10 [2], the hexagrid, diagrid and tube buildings are assumed to design with R values 8, 6 and 8, respectively. The validity of this assumption is verified through collapse-based assessment procedure in this paper. ETABS program [3] is used in the initial analysis and design of the prototypes. For all systems and all of the models, the size of the cross section of the internal members is the same. In fact, the internal members are assumed to carry only gravity loads. The designs satisfy an allowable lateral displacement limit of (H/400). *Table 1* lists the gravity loads for the analytical structures.

Beams, columns and braces have been designed with hollow steel sections with yield stress of 235 N/mm2.

Braces angle for hexagonal system according to previous conclusions [13] and based on the architectural configuration, an angle of 60° has been used for hexagrid structure (θ in *Figure 1*b). Braces angle for diagrid system, based on the architectural configuration, an angle of 41° has been used (θ in *Figure 1*a).

Table 1. *Gravity loads of models*

Description	*Value*
Floor dead load	7.45 kN/m^2
Floor live load	2.0 kN/m^2

Table 2 through *Table 7* shows the frame sizing results of the archetypes that are derived from capacity design approach.

In the following, the building models are appointed by means of an acronym defined by a letter indicating the type of cross section (b for braces, h and v, respectively, for horizontal and vertical rectangular cross section).

Table 2. *Cross sections for 30-story hexagrid system*

Story no.	*Section (a x b x t) [mm x mm x mm]*		
	b	h	v
1 to 10	300 x 300 x 40	300 x 300 x 10	900 x 900 x 40
11 to 20	300 x 300 x 20	300 x 300 x 10	600 x 600 x 30
21 to 30	300 x 300 x 20	300 x 300 x 10	450 x 450 x 30

Table 3. *Cross sections for 30-story diagrid system*

Story no.	*Section (a x b x t) [mm x mm x mm]*	
	b	v
1 to 10	200 x 200 x 20	1200 x 1200 x 60
11 to 20	200 x 200 x 10	800 x 800 x 40
21 to 30	200 x 200 x 10	500 x 500 x 40

Table 4. *Cross sections for 30-story tube system*

Story no.	*Section (a x b x t) [mm x mm x mm]*	
	h	v
1 to 10	300 x 300 x 20	800 x 800 x 40
11 to 20	300 x 300 x 20	300 x 300 x 40
21 to 30	200 x 200 x 20	300 x 300 x 20

Table 5. *Cross sections for 50-story hexagrid system*

Story no.	*Section (a x b x t) [mm x mm x mm]*		
	b	h	v
1 to 10	360 x 360 x 40	300 x 300 x 10	1450 x 1450 x 70
11 to 20	360 x 360 x 40	300 x 300 x 10	1200 x 1200 x 60
21 to 30	300 x 300 x 40	300 x 300 x 10	1100 x 1100 x 40
31 to 40	300 x 300 x 40	300 x 300 x 10	800 x 800 x 40
41 to 50	300 x 300 x 20	300 x 300 x 10	320 x 320 x 40

Table 6. *Cross sections for 50-story diagrid system*

Story no.	*Section (a x b x t) [mm x mm x mm]*	
	b	v
1 to 10	200 x 200 x 20	1700 x 1700 x 90
11 to 20	200 x 200 x 20	1400 x 1400 x 80
21 to 30	200 x 200 x 20	1200 x 1200 x 60
31 to 40	200 x 200 x 20	900 x 900 x 50
41 to 50	200 x 200 x 10	450 x 450 x 40

Table 7. *Cross sections for 50-story tube system*

Story no.	*Section (a x b x t) [mm x mm x mm]*	
	h	v
1 to 10	300 x 300 x 40	900 x 900 x 50
11 to 20	300 x 300 x 40	500 x 500 x 40
21 to 30	300 x 300 x 40	300 x 300 x 40
31 to 40	300 x 300 x 40	300 x 300 x 40
41 to 50	300 x 300 x 40	300 x 300 x 40

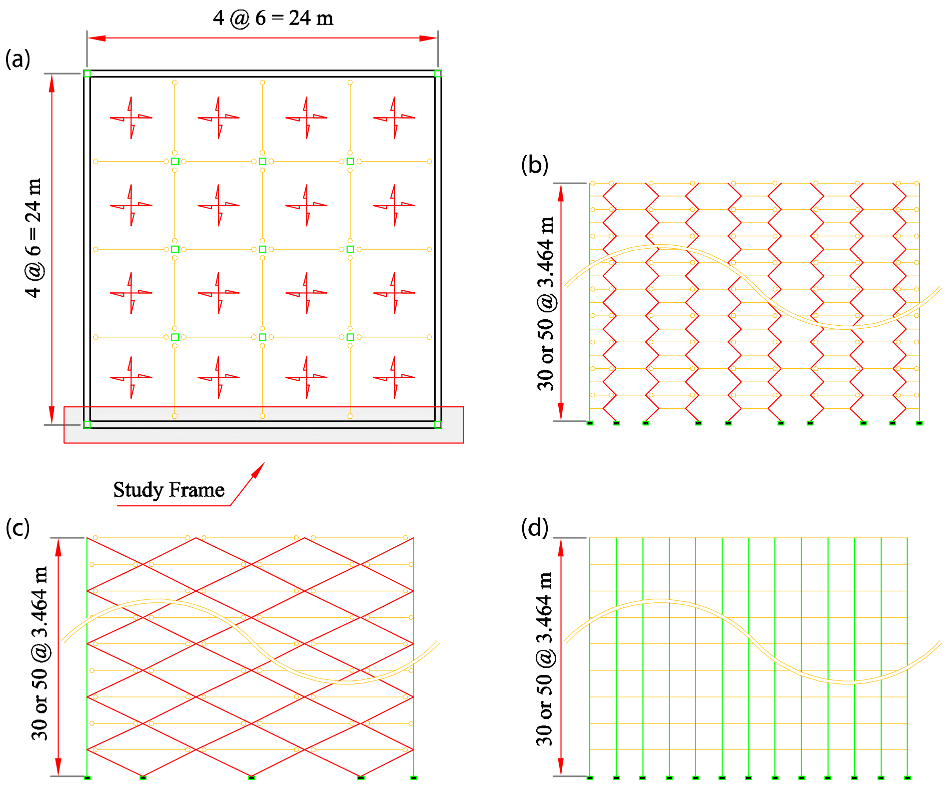

Figure 2. *Steel structure systems: (a) plan; (b) hexagrid elevation; (c) diagrid elevation; (d) tube elevation*

4. STRUCTURAL NONLINEAR MODEL

Nonlinear response history analysis is a robust and effective tool to dynamically determine seismic demands, as well as to identify plastic hinge mechanisms in structures. Conversely, pushover analysis suffers from several inherent deficiencies and limitations [11]. As a result, nonlinear dynamic analysis has currently become a favorite technique for buckling evaluation and design verification of high-rise moment resisting frames (MRFs) [5], with rapid developments in computer technology and computational algorithms.

Tensile yielding and buckling behavior of the brace elements along with local yielding of the beams and columns are the key seismic responses of the previous systems. This paper provides a discrete computational model for collapse assessment of the hexagrid, diagrid and tubular systems using OpenSees [15]. Nonlinear beam-column elements with inelastic fiber sections were used to model all beams and columns. Floor masses were lumped into the column nodes at each story. Material behavior for all beams, columns and EBF braces was modeled using Giuffre-Menegotto-Pinto material model [7] with isotropic hardening and yield strength of 235 N/mm2.

Figure 3 shows a detailed scheme of two-dimensional model of the system. As can be observed, all the prototypes include a leaning column on right side of the model with no lateral stiffness to simulate geometric nonlinearity (P-Δ) effects that are modeled using "elastic Beam-Column", low stiffness "rotational spring" and "rigid truss" elements. Leaning column representing all gravity columns associated with the frames (one half of the building gravity columns were considered). The representative column was pinned at the base, and rigidly constrained to match the frame deformation at each floor.

To simulate post-buckling response of braces, each bracing member is divided into ten segments with shifted nodes perpendicular to the member axis and a value of L/1000 is selected for the initial imperfection at mid-span, where L is the brace length.

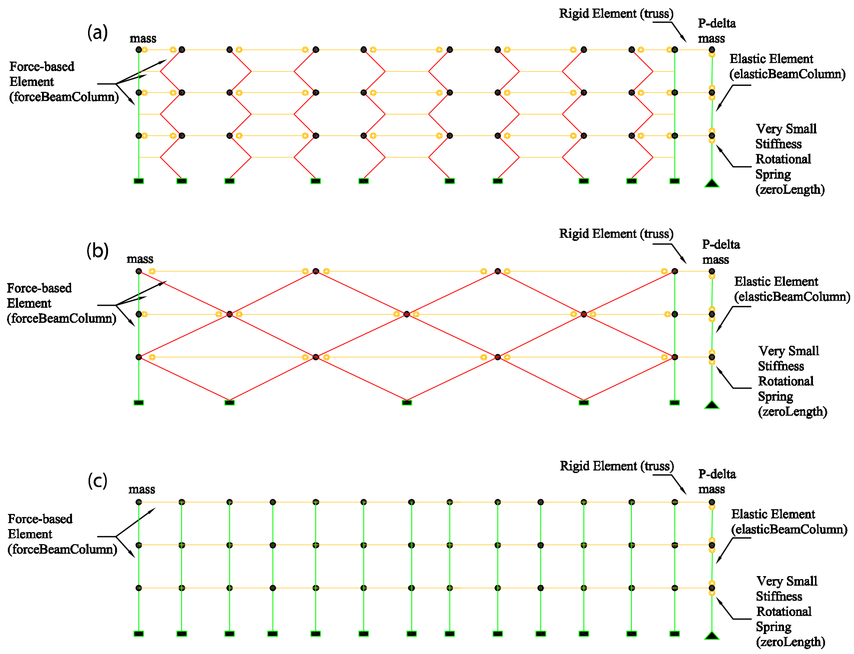

Figure 3. *Nonlinear simulation in Opensees (a) hexagrid system; (b) diagrid system; (c) tube system*

5. NONLINEAR ANALYSES

Previous studies have indicated that building design with proper seismic performance factors, including system overstrength (Ω_0), period-based ductility (μ_T) and response modification coefficient (R), provide a sufficient margin of safety for structural buildings during an earthquake [4]. Current building codes (e.g. [2]) have suggested constant values of seismic performance factors for different code-compliant buildings. FEMA P695 [6] provides a procedure to assess and quantify the seismic performance factors for the existing and new structural buildings through incremental dynamic analysis (IDA) [20].

5.1 Nonlinear Time History Analysis

Nonlinear response of the studied models is evaluated for 7 pairs of far-field ground motion records. These ground motion records have been scaled based on the ground motion scaling requirements of Iran's 2800 seismic provisions [21]. Table 8, represents the properties of the ground motion records used for nonlinear dynamic time history analyses.

Table 8 *Ground motion records used in nonlinear time history analysis*

ID No.	*Earthquake*			*Station*	*Soil Data*	*Site-Source Distance*	*Normalization Factor*
	Magnitude	Year	Name	Name	NEHRP	Joyner Boore	
1	6.61	1971	San Fernando	Fairmount	C	25.6	7.3
2	6.5	1976	Firiuli	Barcis	C	49.1	3.3
3	5.9	1978	Santa Barbara	Cachuma	C	23.7	9.9
4	5.9	1979	Norcia	Bevagana	C	31.4	4.5
5	7.0	1992	Cape Mendocino	Shelter	C	26.5	58.4
6	7.3	1992	Landers	Silent Valley	C	50.8	2.5
7	6.7	1994	Northridge	Alhambra	C	35.7	3

The nonlinear inter-story drift of each studied structures is shown in figures 4 to 9.

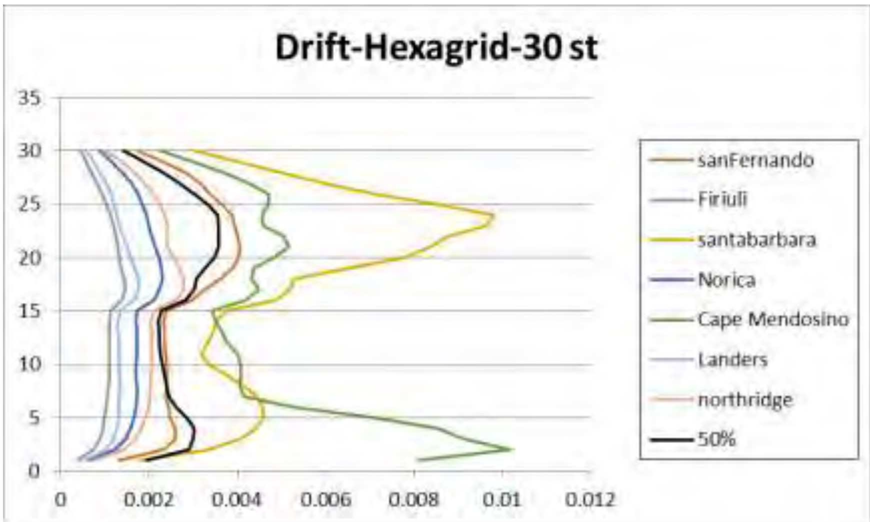

Figure 4 *inter-story drift of 30-story hexagrid structure*

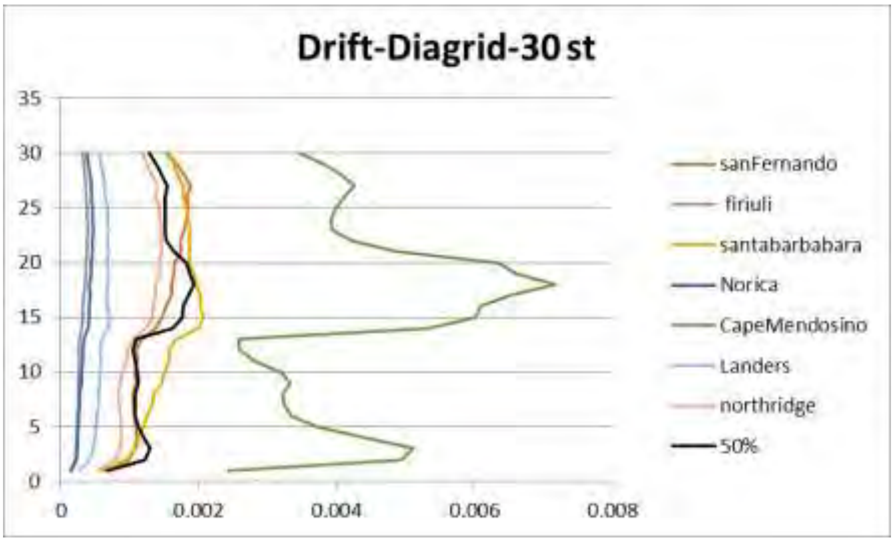

Figure 5 *inter-story drift of 30-story diagrid structure*

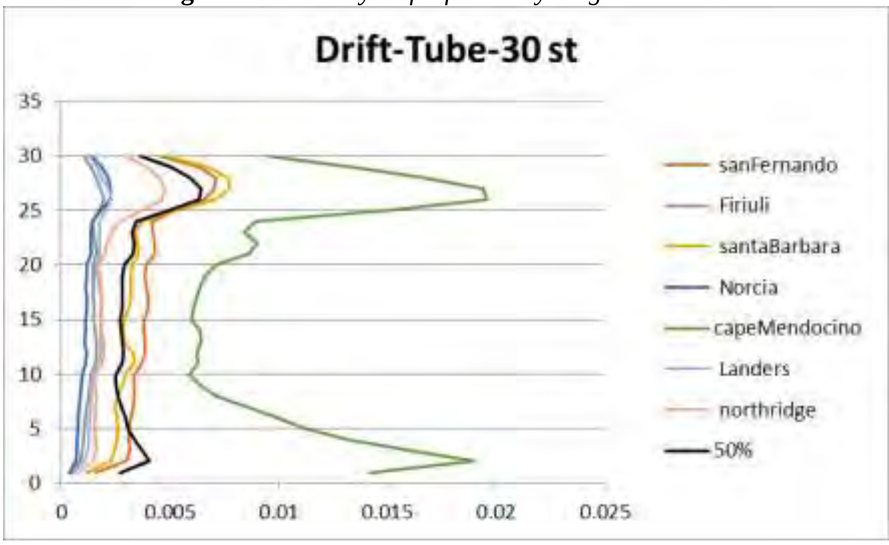

Figure 6 *inter-story drift of 30-story tube structure*

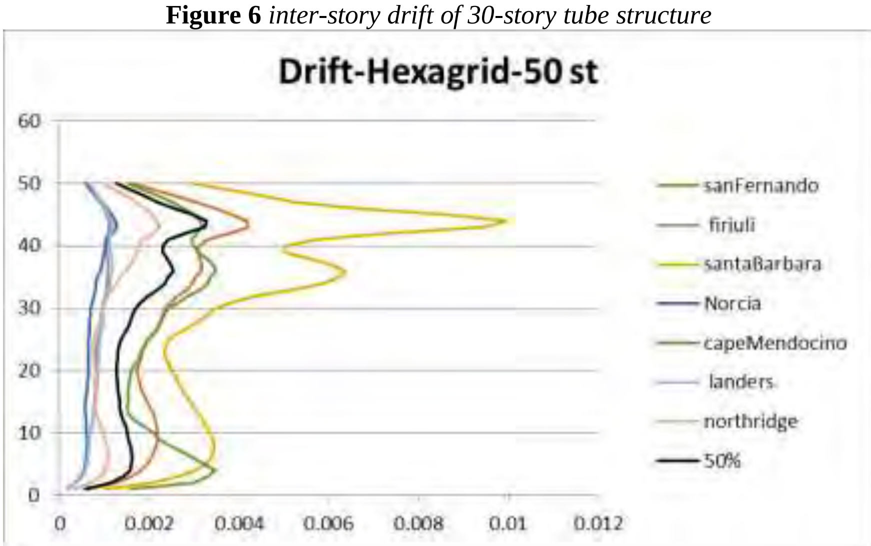

Figure 7 *inter-story drift of 50-story hexagrid structure*

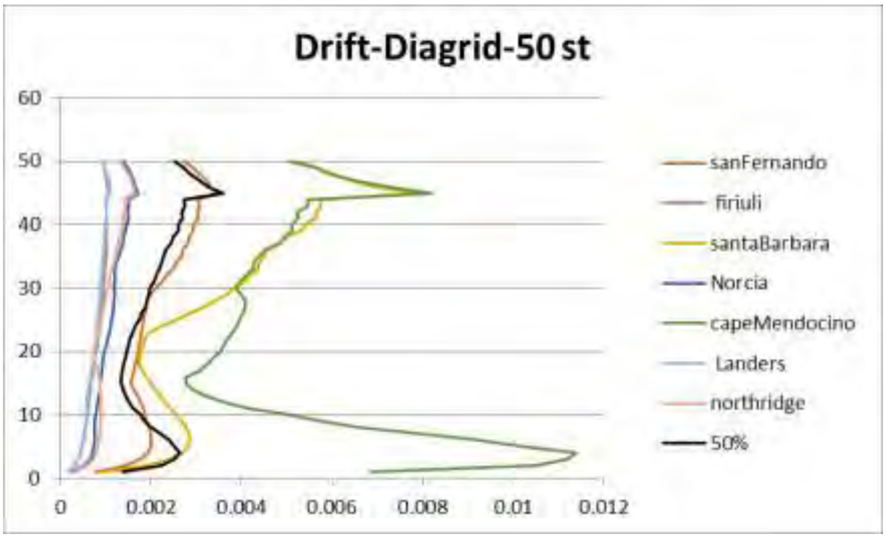

Figure 8 *inter-story drift of 50-story diagrid structure*

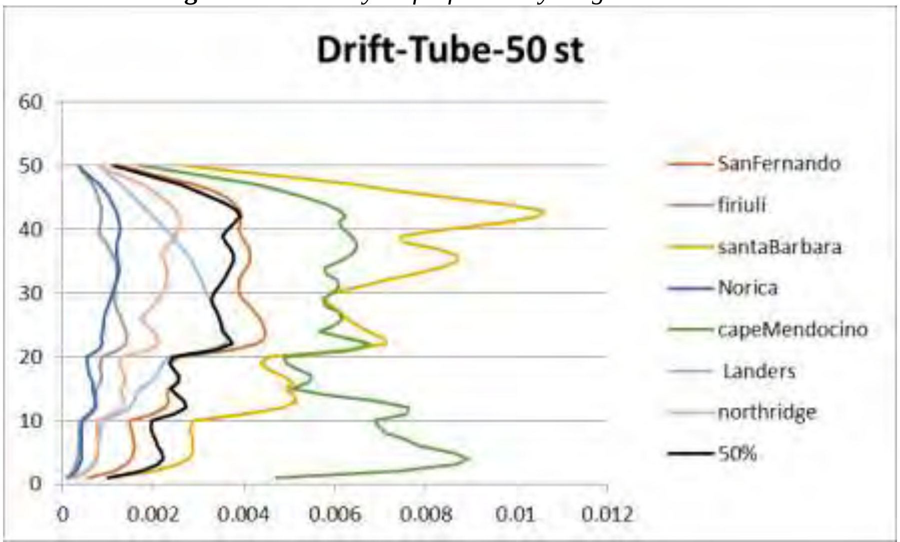

Figure 9 *inter-story drift of 50-story tube structure*

The results obtained from nonlinear time history analyses are summerized in table 9.

Table 9 *Summary of the inter story drift results*

Structural system	Average of Maximum inter-story Drift		Ratio to the Tube system	
	30-story	50-story	30-story	50-story
Hexagrid	0.0037	0.0037	0.61	0.8
Diagrid	0.0018	0.0032	0.3	0.9
Tube	0.006	0.004	1	1

5.2 Nonlinear Static Analysis

In accordance with FEMA P695, values of Ω_0 and μ_T for each building are computed using results of pushover analysis as shown in the following equation [5]:

$$\Omega_0 = \frac{V_{max}}{V} \qquad (1)$$

$$\mu_T = \frac{\delta_u}{\delta_{y,eff}} \qquad (2)$$

Where Vmax and V denote maximum lateral capacity and design base shear of a system, respectively. δu and δy,eff are ultimate roof drift ratio (RDR) corresponding to 0.8 times of Vmax and effectively yield, respectively (as shown in *Figure 10*).

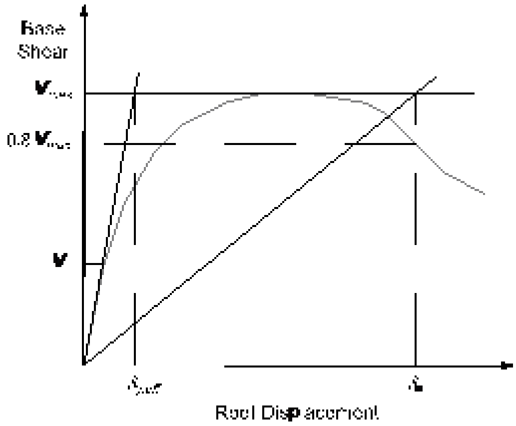

***Figure 10.** Idealized nonlinear static pushover curve [6]*

Figure 11 and Figure 12 show the pushover curve of hexagrid, diagrid and tube system of 30- and 50-story, respectively. As shown, Ω_0 of the buildings is defined as the ratio of the maximum base shear, V_{max}, to the design base shear, V, and μ_T, is the ratio of ultimate, δ_u, to effective yield RDR, $\delta_{y,eff}$.

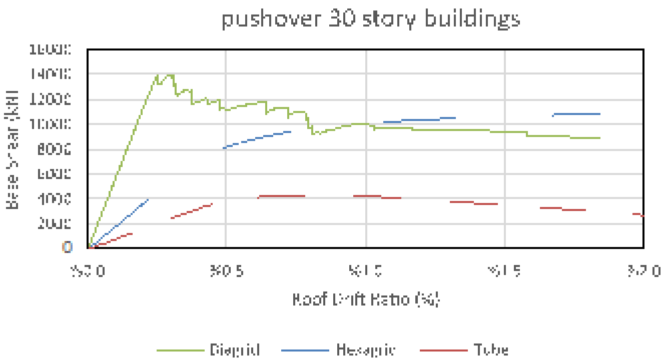

***Figure 11.** Pushover curve of 30-story buildings*

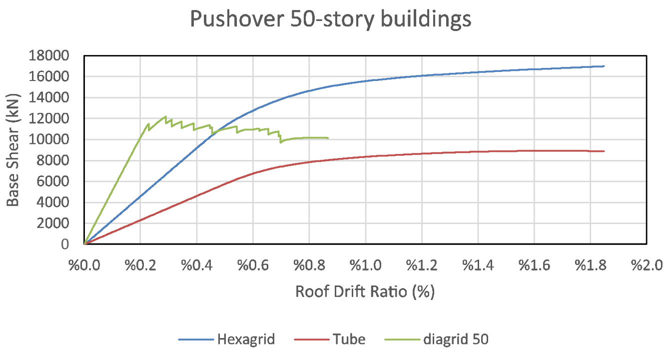

***Figure 12.** Pushover curve of 50-story buildings*

Summary of pushover results for all the buildings is listed in Table 10.

***Table 10.** Summary of pushover results*

Specimen	Number of story	Pushover results	
		Ω_0	μ_T
Hexagrid	30	1.2	2.7
Diagrid	30	1.6	2.6
Tube	30	1.0	2.2
Hexagrid	50	1.5	3.8
Diagrid	50	1.4	2.5
Tube	50	1.3	2.0

5.3 Nonlinear Incremental Dynamic Analysis

To assess the validity of the proposed response modification coefficient (R) for structural systems, it is required to perform incremental dynamic analysis (IDA). IDA is an analytical method for estimating the distribution of demand and capacity of the structure under predefined multiply scaled records, in which a series of nonlinear dynamic analysis is performed from low to collapse intensities [19]. In this paper, IDA is carried out using near-field ground motion set, composed of 10 component pairs of horizontal ground motions. The records are listed in Table 11, and Figure 13 shows the spectral acceleration of the unscaled near-field record set.

Table 11 *Dataset of the near-field ground motion records*

ID	Event	Year	Station	M_w	R_{jb} (km)	PGA_{max} (g)
1	Irpinia Italy	1980	Bagnoli Irpinio	6.90	8.14	0.19
2	Morgan Hill	1984	Gilroy Array #6	6.19	9.85	0.29
3	Loma Prieta	1989	Saratoga – Aloha Ave	6.93	7.58	0.51
4	Cape Mendocino	1992	Petrolia	7.01	0	0.66
5	Northridge-01	1994	Jensen Filter Plant	6.69	0	1.00
6	Northridge-01	1994	LA Dam	6.69	0	0.43
7	Northridge-01	1994	Pacoima Kagel Canyon	6.69	5.26	0.43
8	Northridge-01	1994	Sylmar – Converter Sta	6.69	0	0.85
9	Northridge-01	1994	Sylmar – Olive View	6.69	1.74	0.85
10	Chi-Chi Taiwan	1999	TCU068	7.62	0	0.51

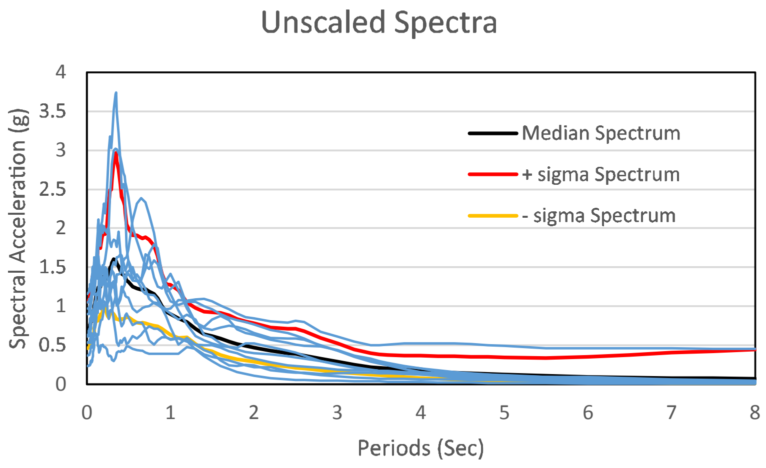

Figure 13. *Five percent damped spectral acceleration of the near-field record set*

6. PERFORMANCE EVALUATION

Validity of the trial response modification coefficient, R, used for the design of hexagrid, diagrid and tube buildings is evaluated using the FEMA P695 procedure. For this purpose, to examine the validity of this assumption, collapse margin ratios (CMRs) of this buildings are adjusted with total uncertainty and spectral shape effects, and the adjusted CMRs are compared with the acceptable criteria, as explained further blow:

In accordance with FEMA P695, CMR is first obtained from IDA curves by the following equation:

$$CMR = \frac{\hat{S}_{CT}}{S_{MT}} \tag{3}$$

Where $\hat{S}_{CT}$ is the spectral intensity of median collapse capacities, which corresponds to 50% probability of collapse, and S_{MT} denotes the spectral intensity of maximum considered earthquake (MCE) spectrum at the fundamental period of the buildings. Figure 14 through Figure 19 shows IDA curves of 30- and 50-story, hexagrid, diagrid and tube system, respectively. It shows that inter-story drift of hexagrid system reached 10% or "collapse prevention" limit state in higher spectrum acceleration than diagrid and tube systems.

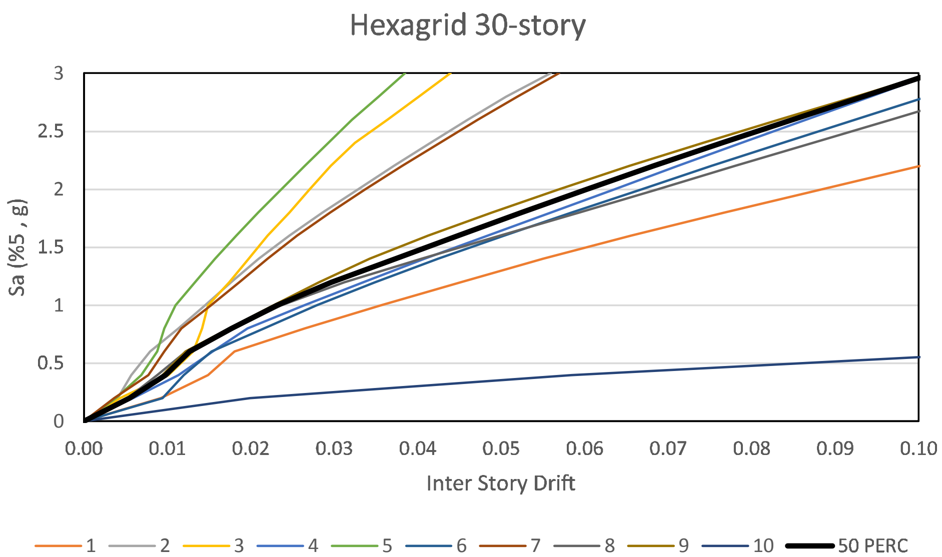

Figure 14. *IDA results of 30-story hexagrid system*

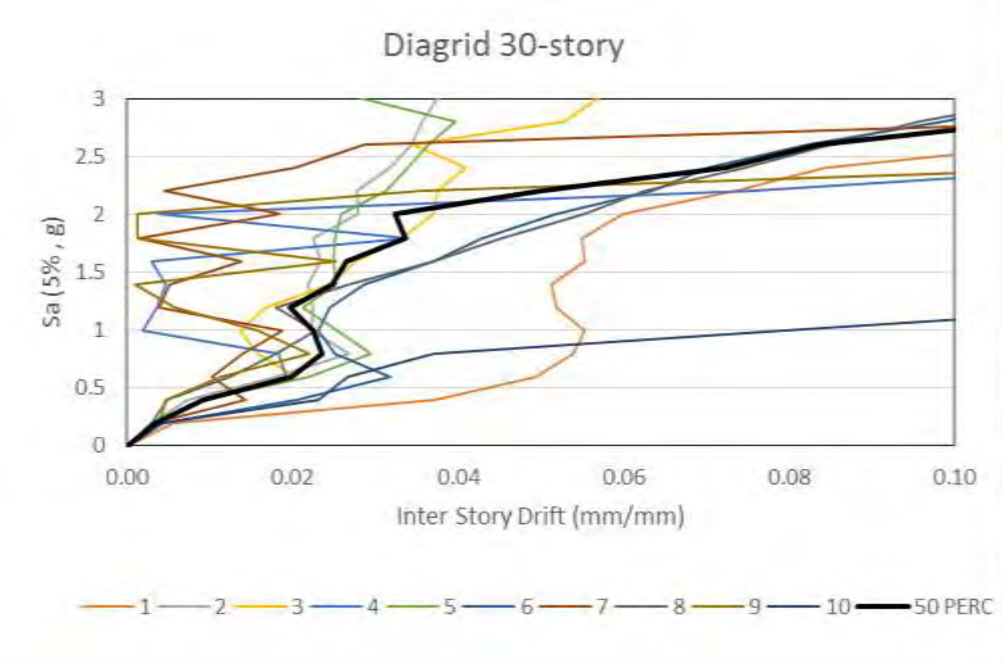

Figure 15. *IDA results of 30-story diagrid system*

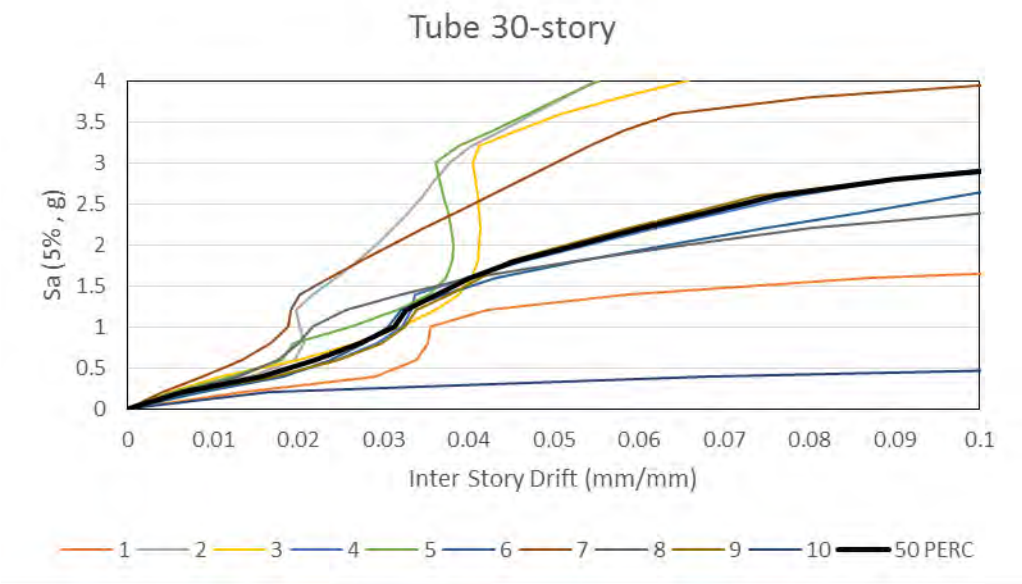

Figure 16. *IDA results of 30-story tube system*

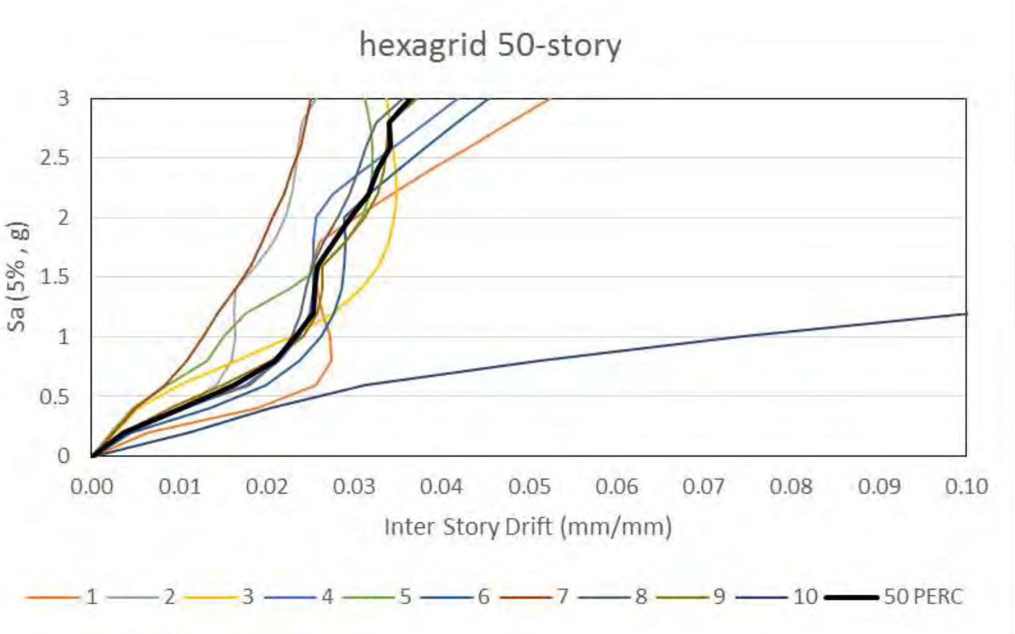

Figure 17. *IDA results of 50-story hexagrid system*

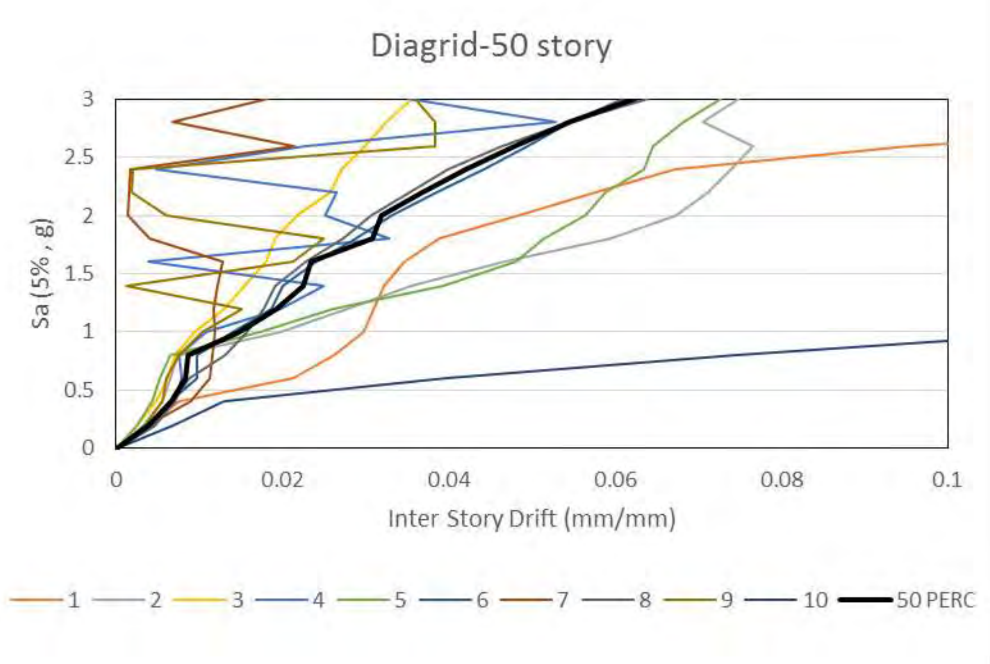

Figure 18. *IDA results of 50-story diagrid system*

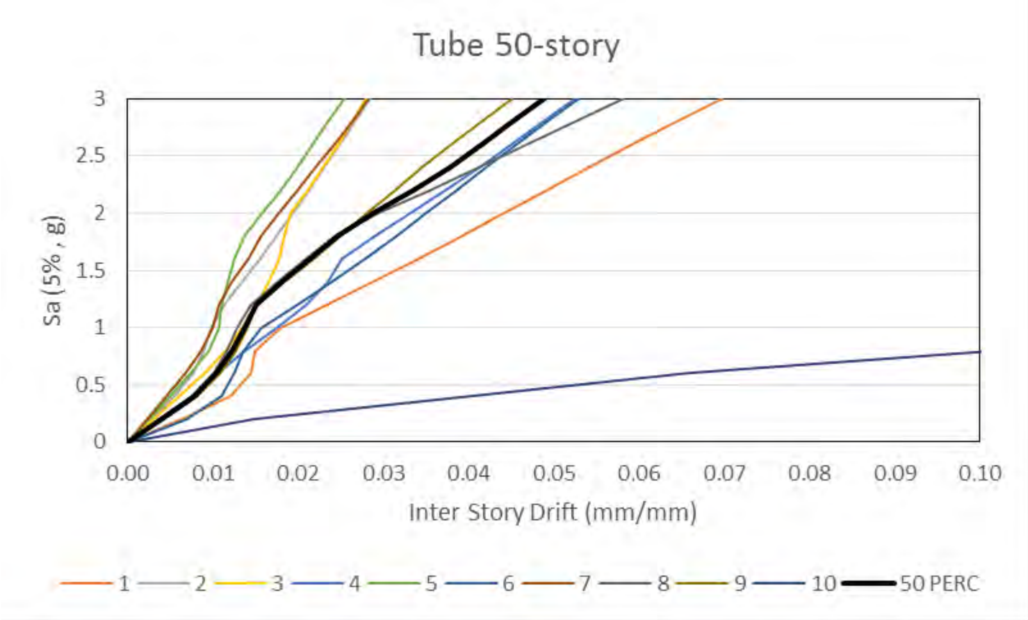

***Figure 19.** IDA results of 50-story tube system*

CONCLUSION

This paper investigated the quantification of three seismic performance factors, including over-strength factor (Ω_0), period-based ductility (μ_T) and response modification coefficient (R) for hexagrid, diagrid and tube systems. Nonlinear static analysis was conducted to obtain seismic performance factors of over-strength (Ω_0) and period-based ductility (μ_T). Incremental dynamic analyses were carried out to evaluate the validity of the trial response modification coefficient (R) by a collapse assessment procedure. Also, nonlinear time history analyses were conducted to obtain the nonlinear inter-story drift of the studied models. The investigation resulted in the following main conclusions:

1- Diagrid systems are stiffer than hexagrid systems and they are stiffer than tube systems.

2- Period based ductility (μ_T) are ranged from 2.0 to 3.8, which ductility of hexagrid system is 3.8 which shows that this system has higher ductility than diagrid and tube systems. Tube systems has minimum ductility which shows them worse than hexagrid and diagrid systems.

3- Over-strength factor (Ω_0) are ranged from 1.0 to 1.6, which over-strength factor of diagrid system is 1.6. 50-story hexagrid system shows most over-strength factor among 50-story buildings that makes it safe during earthquake. Tube system with lowest over-strength factor shows weak performance.

4- However, Diagrid system are stiffer than hexagrid system, but spectral acceleration according to “limit state” of hexagrid system is higher than diagrid system that shows the better performance of hexagrid system.

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